

ECC

Alex Chai

Anthony Chang

Erin Jaen

Henry Yuen

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Chapter 1

Basics of Codes

1.1 Definitions

Definition 1 (Code). A code C of block length n over an alphabet α is a subset of α^n . Typically, we will use q to denote the alphabet size $|\Sigma|$.

Definition 2 (Alphabet Size). The alphabet size is $q := |\alpha|$, the cardinality of the finite alphabet α .

Definition 3 (Dimension). The dimension k of a code C is

$$k := \log_q(|C|),$$

where q is the alphabet size and $|C|$ is the number of codewords.

Definition 4 (Rate). The rate R of a code C is

$$R := \frac{k}{n}.$$

where k is the dimension and block length of n .

1.2 Properties

Lemma 5 (Dimension is at most the block length). *For every code C of block length n , we have $k \leq n$.*

Proof. Note that $C \subseteq \alpha^n$, we have $|C| \leq |\alpha|^n = q^n$, thus $k = \log_q |C| \leq \log_q q^n = n$. □

Lemma 6 (Rate is at most one). *For every code C , we have $R \leq 1$.*

Proof. By Lemma 5, $k \leq n$, and $n \geq 0$. Therefore $R = k/n \leq 1$. □

Chapter 2

Linear Codes

2.1 Definitions

Definition 7 (Linear Code by Subspace). Let q be a prime power (i.e. $q = p^s$ for some prime p and integer $s \geq 1$). $C \subseteq \mathbb{F}_q^n$ is a linear code if it is a linear subspace of $\mathbb{F}_q^n$. If C has dimension k and distance d then it will be referred to as an $[n, k, d]_q$ or just an $[n, k]_q$ code.